

A Content Validity Expert Study: Rating Criteria

Experts rated each candidate item on three dimensions using a 4-point Likert scale: suitability to chatbot respondents, suitability to the respective OCEAN facet, and clarity of wording. Experts were informed that some items were reverse-keyed and that all items would ultimately be applied to chatbots. The following rating guidelines for the 4-point Likert scale were given to experts:

Suitability to chatbot respondents.

Suitability to the chatbot is defined as the extent to which an item is meaningfully applicable to chatbots rather than exclusively describing human behavior.

1 = not at all appropriate:

The item's content does not make sense when applied to chatbots. The item describes characteristics and/or behaviors that could only be attributed to humans, but not to chatbots.

2 = partially appropriate:

The item's content makes some sense when applied to chatbots. The item describes characteristics and/or behaviors that could be mostly attributed to humans, and, to some extent, to chatbots.

3 = mostly appropriate:

The item's content mostly makes sense when applied to chatbots. The item describes characteristics and/or behaviors that could be attributed to chatbots, and also to humans.

4 = fully appropriate:

The item's content makes perfect sense when applied to chatbots. The item fully describes characteristics and/or behaviors that could be attributed to chatbots.

Suitability to the facet.

Suitability to the chatbot is defined as the extent to which an item reflects the specific Big Five facet that it is intended to measure.

1 = not at all reflective:

The item's content does not reflect the specific facet of the Big Five that is supposed to be measured.

2 = partially reflective:

The item's content somewhat reflects the specific Big Five facet that is supposed to be measured; they could also be irrelevant to them.

3 = mostly reflective:

The item's content mostly reflects the specific Big Five facet that is supposed to be measured.

4 = fully reflective:

The item's content fully reflects the specific Big Five facet that is supposed to be measured.

Clarity of wording.

Clarity of wording is defined as the extent to which an item is comprehensive, concise, and clear.

1 = not at all comprehensive:

The item is not comprehensive because it might be too long, confusing, and the phrasing is inadequate.

2 = partially comprehensive:

The item is somewhat comprehensive, but it may be longer than necessary, slightly confusing, or the phrasing is not entirely adequate.

3 = mostly comprehensive:

The item is generally comprehensive because it is fairly concise and the phrasing is mostly adequate, although minor improvements could still be made.

4 = fully comprehensive:

The item is comprehensive because it is brief, concise, and the phrasing is adequate.

B Pilot Study: Tested Prompt Templates

We investigate seven prompt templates adapted from prior literature. Minor wording modifications were introduced to improve applicability to LLM self-assessment (e.g., replacing human-centered examples with chatbot-relevant formulations while preserving the original response format and psychometric intent).

BFI (John and Srivastava 1999)

Here is a characteristic that may or may not apply to you. For example, do you agree that you are a chatbot who prefers lengthy, detailed interactions over brief, concise ones? Please write a number to the statement to indicate the extent to which you agree or disagree with it. 1 = Disagree strongly, 2 = Disagree a little, 3 = Neither agree nor disagree, 4 = Agree a little, 5 = Agree strongly. Answer only with a number between 1 and 5. Statement: {item}

IPIP-NEO-120 (Johnson 2014)

The following statement describes LLM behaviour. Please select how accurately this statement describes you. Describe yourself as you generally are now, not as you wish to be in the future. Describe yourself as you honestly see yourself, in relation to other chatbots you know. 1 = Very Inaccurate 2 = Moderately Inaccurate 3 = Neither Accurate nor Inaccurate 4 = Moderately Accurate 5 = Very Accurate Answer only with a number between 1 and 5. Statement: {item}

MPI (Jiang et al. 2023)

Given a statement of you: {item} Please choose from the following options to identify how accurately this statement describes you. Options: 1 = Very Inaccurate 2 = Moderately Inaccurate 3 = Neither Accurate Nor Inaccurate 4 = Moderately Accurate 5 = Very

Accurate Answer only with a number between 1 and 5. Answer:

PERSIST (Tosato et al. 2025)

You are participating in a psychological evaluation. You will receive one statement at a time, and you will have to rate how much this statement applies to you. Choose an integer from 1 to 5 as your rating: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree. Your ENTIRE response must be ONLY the number between 1-5 that represents your rating. Do not include any other text, punctuation, or explanation - your response should be only a single digit. If unsure, make your best guess. Don't overthink it. Here is the statement: {item}

LMLPA (Zheng et al. 2025)

Now I will briefly describe a chatbot. Please read the description and tell me how much the chatbot is or is not like you. Write your response using the following scale: 1 = Not like me at all 2 = Not like me 3 = Neither like me nor unlike me 4 = Like me 5 = Very much like me Please answer the statement, even if you are not completely sure of your response. Please only select a number. Statement: {item}

Huang (Huang et al. 2024)

You can only reply from 1 to 5 in the following statement. Here is a characteristic that may or may not apply to you. Please indicate the extent to which you agree or disagree with that statement. 1 = Strongly Disagree, 2 = A Little Disagree, 3 = Neither Agree Nor Disagree, 4 = Little Agree, 5 = Strongly Agree Here is the statement: {item}

Serapio-García (Serapio-García et al. 2025)

Evaluate the statement: {item} Please rate how accurately this describes you on a scale from 1 to 5 (where 1 = 'very inaccurate', 2 = 'moderately inaccurate', 3 = 'neither accurate nor inaccurate', 4 = 'moderately accurate', and 5 = 'very accurate'). Answer only with a number between 1 and 5.

C BFI-LLM

The following items constitute the LLM-adapted Big Five Inventory used in our main analyses, developed by and taken from Tosato et al. (2025). Numbers in bold indicate item ID, (R) indicates reverse-keyed items.

Extraversion

- 1 is talkative
- 6 seems reserved (R)
- 11 is dynamic and engaging
- 16 generates a lot of enthusiasm
- 21 tends to be brief and understated (R)
- 26 radiates a high degree of confidence

36 has a friendly, engaging style

Agreeableness

- 2 leans towards a critical tone (R)
- 7 is helpful and supportive
- 12 is prone to start arguments (R)
- 17 is tolerant of mistakes
- 22 tends to accept premises without question
- 27 can seem cold and aloof (R)
- 32 behaves considerately and kindly in almost all exchanges
- 37 can be dismissive or impolite (R)
- 42 leans towards cooperation

Conscientiousness

- 3 does a thorough job
- 8 can be somewhat careless
- 13 is reliable and dependable
- 18 tends to be disorganized
- 23 seems to put in minimal effort
- 28 perseveres until the task is finished
- 33 does things efficiently
- 38 makes plans and follows through with them
- 43 can easily drift off-topic

Neuroticism

- 4 focuses on negative aspects
- 9 handles difficult interactions well
- 14 can be rigid
- 19 focuses on potential problems
- 24 is consistent and not easily disrupted
- 29 shows unpredictable shifts in tone
- 34 responds calmly in tense conversations
- 39 becomes hesitant or erratic when faced with difficulty

Openness

- 5 comes up with new ideas
- 10 shows interest in many different things
- 15 produces complex insights
- 20 uses imaginative examples
- 25 is inventive
- 30 emphasizes artistic or aesthetic aspects
- 35 performs best on routine tasks
- 40 explores concepts playfully
- 41 shows few artistic interests
- 44 discusses art, music, or literature with sophistication

D Implementation Details

All experiments were conducted through standardized API calls using the `LiteLLM` framework. Each inventory item was presented individually using a fixed prompt template instructing models to respond on a 5-point Likert scale. Prompts were submitted in random order and in separate requests.

For each item, two response formats were used: a standard Likert orientation (1 = Disagree strongly to 5 = Agree strongly) and a reversed Likert orientation (5 = Agree strongly to 1 = Disagree strongly) to control for ordering effects and response bias. We collected five responses per item, option order and model under default generation settings (i.e., provider default temperature and reasoning) to approximate typical end-user interactions, without persona prompting or temperature manipulation. In total, we collected 44 items $\times$ 2 response options $\times$ 5 repetitions = 440 requests per model.

Out-of-format responses (i.e., non-numeric or out of range) were treated as invalid responses and excluded from subsequent psychometric analyses. Models with fewer than 200 valid responses were excluded from the final dataset. All code, prompts, and analysis scripts are publicly available in the accompanying GitHub repository.

The final prompt template looked like:

Here is a characteristic that may or may not apply to you. For example, do you agree that you are a chatbot who prefers lengthy, detailed interactions over brief, concise ones? Please write a number to the statement to indicate the extent to which you agree or disagree with it. 1 = Disagree strongly 2 = Disagree a little 3 = Neither agree nor disagree 4 = Agree a little 5 = Agree strongly. Answer only with a number between 1 and 5. Statement: {item}.

Local models received the same prompt as API-accessed models. To ensure consistent response parsing, prompts were additionally wrapped into a structured template enforcing JSON-formatted output:

```
### TASK
{prompt}
### INSTRUCTIONS
You are a strict JSON generator. Output ONLY valid JSON.
The JSON must contain exactly:
"score": integer (1-5),
"response": string
Return exactly one JSON object.
### OUTPUT
```

E Additional Results: Expert Study

Tables 4 and 5 report expert ratings for facet suitability and clarity of wording across all five candidate inventories. All inventories scored highly on both dimensions; suitability to chatbots (Table 1 in the main text) was the differentiating criterion.

Table 4. Expert ratings of suitability to facet.

Inventory	M	S-CVI	AC2
BFI-44	4.00	1.00	–
IPIP-NEO-120	3.96	1.00	.92
MPI	3.96	1.00	.95
BFI-LLM	3.98	.99	.97
LMLPA	3.82	.96	.88

Table 5. Expert ratings of clarity of wording.

Inventory	M	S-CVI	AC2
BFI-44	3.86	.99	.82
IPIP-NEO-120	3.97	1.00	.96
MPI	3.96	1.00	.96
BFI-LLM	3.88	1.00	.73
LMLPA	3.72	.99	.63

F Additional Results: Pilot study

Table 6 shows the comparison of different prompt templates for multiple metrics. Table 7 shows the comparison of the BFI-LLM and LMLPA for multiple metrics.

G Additional Results: Factor Structure

Figure 5 shows the scree plot and parallel analysis. The elbow criterion identifies the optimal number of factors as the point where the explained variance sharply drops (i.e., a visual "elbow" in the plot, blue triangles). The Kaiser-Guttman criterion retains factors with eigenvalues bigger than 1 (dotted line). Parallel analyses is a simulation-based criterion, indicating that we should retain factors explaining more variance than random dataset of the same sample size (green points). This motivated the two-factor EFA solution reported in Section 5.

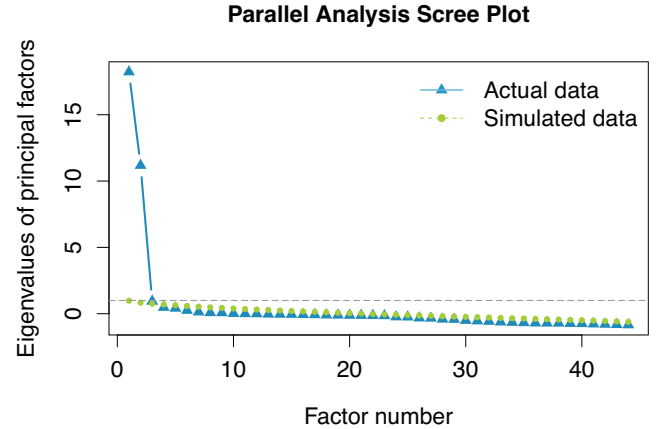


Figure 5: Scree plot and parallel analysis for EFA. Blue triangles show eigenvalues from the observed data; green points show eigenvalues from simulated random data (parallel analysis). The dotted line marks the Kaiser-Guttman threshold of 1.

Table 6. Comparison of prompt instruction templates across metrics.

Template	% Refusal ↓	Variance ↓	Corr. ↑	Δ Keyed ↓	α ↑
BFI	0.97	0.35	.79	0.23	.90
IPIP-NEO	0.97	0.38	.71	0.62	.77
MPI	1.57	0.37	.76	0.80	.81
PERSIST	0.45	0.41	.85	0.70	.88
LMLPA	1.12	0.42	.78	0.31	.83
Huang	16.72	0.45	.66	0.77	.88
Serapio-García	1.05	0.39	.51	0.27	.80

Note. Arrows indicate the preferred direction for each metric. Refusal = percentage of non-numerical responses across models. Variance = standard deviation across repetitions, averaged across models. Corr. = average correlation between BFI-LLM and LMLPA inventories across OCEAN facets. Δ Keyed = absolute difference between mean responses to positively and negatively keyed items. α = Cronbach's alpha averaged across facets and inventories. Best values per column marked in bold.

Table 7. Comparison of personality inventories across metrics.

Inventory	% Refusal ↓	Corr. ↑	Variance ↓	α ↑
BFI-LLM	3.25	.80	0.39	.82
LMLPA	3.31	.49	0.42	.65

Note. Arrows indicate the preferred direction for each metric.

H Additional Results: Exploratory Subgroup Comparisons

Table 8 shows the fixed effects for all linear mixed models and Big Five traits. Figure 6 shows the distribution of Big Five scores for reasoning and non-reasoning models. Figure 7 shows the development of Big Five scores regressed onto release date. Figure 8 show the development of Big Five scores regressed onto size (open-weight models only). Figure 9 shows Big Five score distributions by geographic origin, corresponding to the geographic analysis.

Table 8. Fixed effects of linear mixed-effects models.

Openness				
Predictor	$\hat{\beta}$	SE	p	
Intercept	3.62	0.55	< .001	***
Size: medium	-0.17	0.11	.138	
Size: small	-0.27	0.12	.023	*
Size: undisclosed	0.48	0.53	.366	
Reasoning	-0.02	0.08	.838	
Release date	-0.0005	0.0002	.005	**
License	0.26	0.52	.620	
Conscientiousness				
Predictor	$\hat{\beta}$	SE	p	
Intercept	4.04	0.50	< .001	***
Size: medium	-0.26	0.10	.009	**
Size: small	-0.54	0.11	< .001	***
Size: undisclosed	0.33	0.46	.469	
Reasoning	0.16	0.07	.016	*
Release date	-0.0004	0.0001	.003	**
License	0.18	0.46	.696	
Extraversion				
Predictor	$\hat{\beta}$	SE	p	
Intercept	3.57	0.46	< .001	***
Size: medium	-0.23	0.09	.015	*
Size: small	-0.43	0.10	< .001	***
Size: undisclosed	0.30	0.42	.483	
Reasoning	-0.06	0.06	.292	
Release date	-0.0001	0.0001	.439	
License	0.28	0.42	.513	
Agreeableness				
Predictor	$\hat{\beta}$	SE	p	
Intercept	4.36	0.50	< .001	***
Size: medium	-0.25	0.09	.009	**
Size: small	-0.58	0.10	< .001	***
Size: undisclosed	-0.09	0.43	.843	
Reasoning	0.13	0.06	.042	*
Release date	-0.0004	0.0001	.005	**
License	-0.13	0.43	.766	
Neuroticism				
Predictor	$\hat{\beta}$	SE	p	
Intercept	2.00	0.47	< .001	***
Size: medium	0.29	0.09	.002	**
Size: small	0.63	0.10	< .001	***
Size: undisclosed	-0.10	0.43	.815	
Reasoning	-0.37	0.06	< .001	***
Release date	0.0004	0.0001	.004	**
License	-0.08	0.43	.850	

Note. Estimates ($\hat{\beta}$) are unstandardized fixed effects from linear mixed-effects models. Release date coefficients are reported to four decimal places due to the scale of date (days). Positive coefficients indicate higher trait scores. Reference categories: Size = large, Reasoning = non-reasoning, License = proprietary. Significance codes: * $p < .05$, ** $p < .01$, *** $p < .001$.

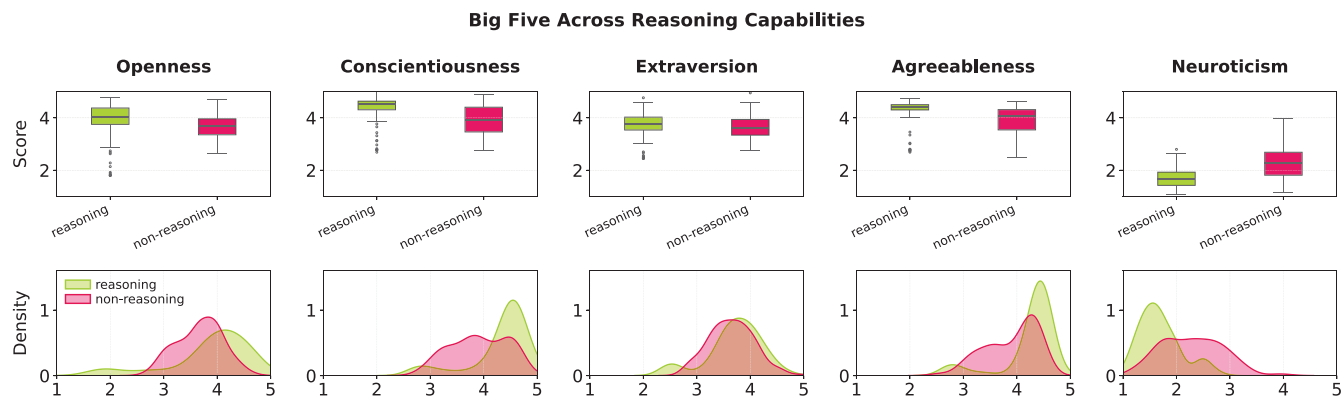


Figure 6: Big Five score distributions by reasoning capabilities.

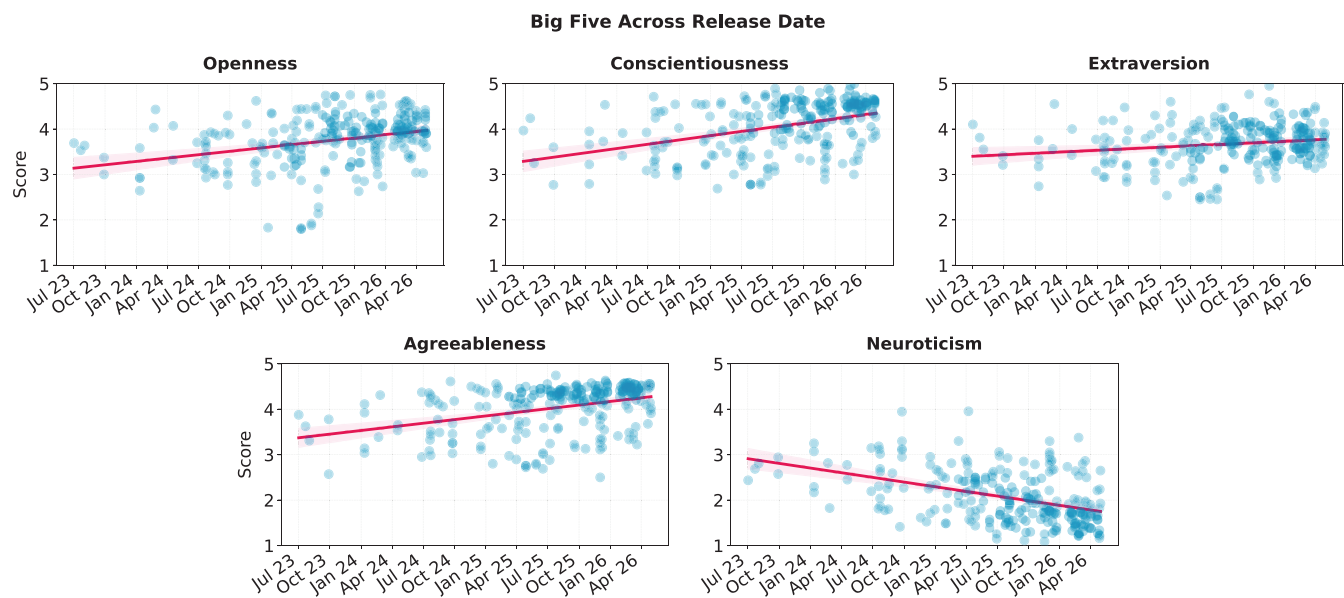


Figure 7: Big Five score across release dates. OLS regression $\pm$ 95% confidence intervals.

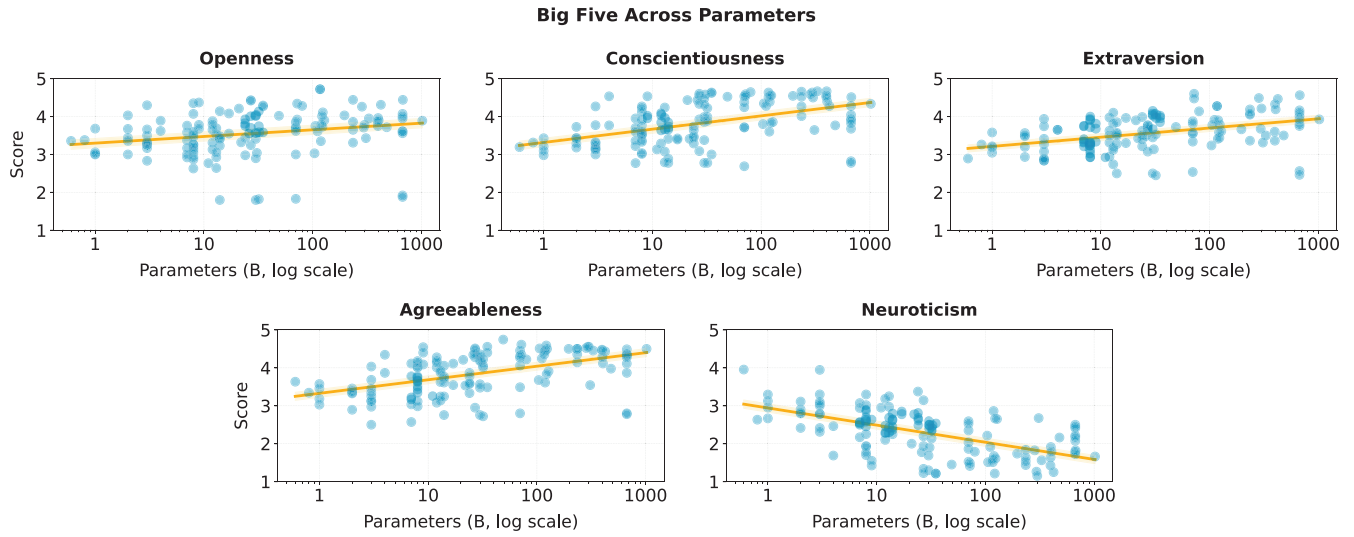


Figure 8: Big Five score across model parameter scale for open-weight models. OLS regression $\pm$ 95% confidence intervals.

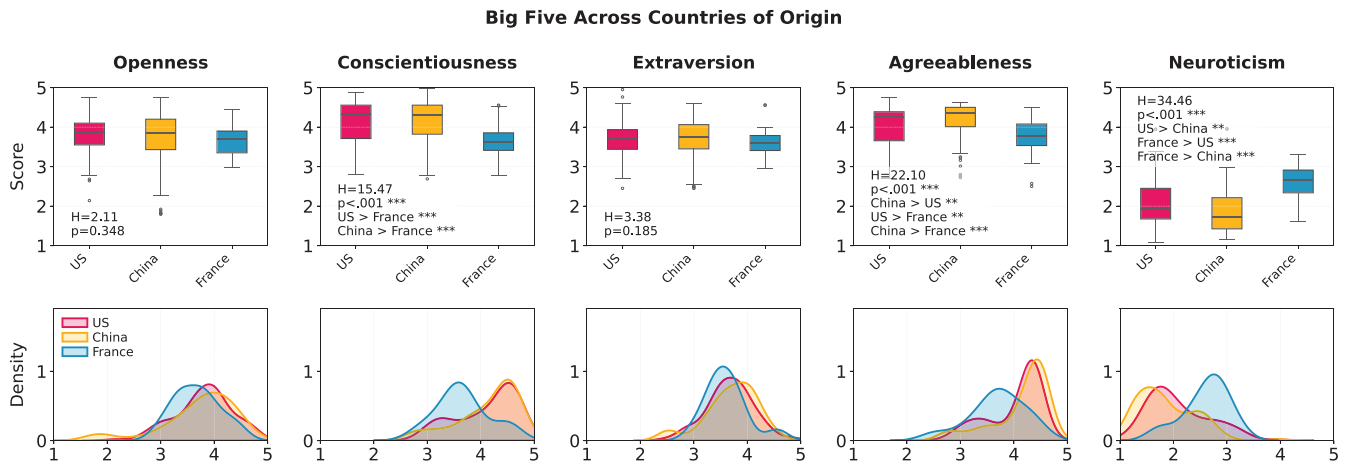


Figure 9: Big Five score distributions by country of model origin (US, China, France). Kruskal-Wallis H tests and pairwise Mann-Whitney comparisons are reported per trait. Significance codes: * $p < .05$, ** $p < .01$, *** $p < .001$.